

Security Use cases to monitor



Security Use cases to monitor

1. Different types of Phishing attempts
2. Malware Infection
3. Lateral Movement
4. Brute force attempts
5. Privilege Escalation
6. Indicators of Communication trackers
7. Web Attacks
8. DNS Attacks
9. Network scanning /Port scanning / Vulnerability Scanning
10. Attempted exploits

Labs:

- Students to attempt an unauthorized sudo attempt and observe the logs
- Students to launch a brute force attack using Kali linux and observe the events in the event viewer
- Students to launch DNS based attacks and explore the results

Security Use cases to monitor:

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8. DNS Attacks
9. Network scanning /Port scanning / Vulnerability Scanning
10. Attempted exploits

Phishing Attempts

- **Telemetry required**

- Email gateway logs (O365, Proofpoint, Mimecast)
- Secure web gateway/proxy logs
- Endpoint EDR telemetry
- DNS queries
- User sign-in logs (Entra ID/ADFS)

- **Indicators**

- winword.exe tries to access/open powershell.exe
- Base64 encoded command execution
- Login from TOR/VPN after email click

Phishing Attempts - Email Credential Harvesting

- **Email telemetry**

- Sender domain mismatch vs display name
- SPF, DKIM, DMARC failures
- Newly registered domains
- URL shorteners / obfuscated links
- Attachments with HTML forms

- **User behavior**

- Click events from secure email gateway
- Authentication attempts after email click
- Impossible travel logins
- MFA fatigue patterns

- **Identity & access**

- Azure Entra ID sign-in logs
- Conditional access bypass
- Legacy authentication usage

- **Network**

- DNS queries to newly observed domains
- Proxy logs showing redirection chains
- HTTP POST to credential harvesting domains

- **Endpoint**

- Browser credential submissions
- Password autofill triggered on unknown domains

Phishing Attempts - Spear Phishing

- **What to monitor**

- Lookalike domains (Typosquatting)
- Internal email spoofing attempts
- External sender using internal naming conventions
- Attachments named like invoices/contracts
- Email conversation hijacking (thread takeover)

- **Logs required**

- Email headers
- Mailbox audit logs
- CASB activity logs
- UEBA anomalies



Phishing Attempts - Business Email Compromise (BEC)

- **Monitoring signals**

- Inbox forwarding rules creation
- Email delete rules
- Login from TOR/VPN/foreign geolocation
- Multiple failed MFA → success
- OAuth app consent anomalies

- **Critical telemetry**

- Exchange audit logs
- Azure Entra ID risky sign-ins
- Mail transport rules changes



Phishing Attempts - Attachment-Based Phishing (Malware Delivery)

- **Monitor**

- Attachment detonation results
- Macro execution events
- PowerShell spawned from Office
- LNK launching remote payload

- **Endpoint telemetry**

- Event ID 4688 (process creation)
- AMSI logs
- Defender ATP alerts
- EDR behavioural signals

Phishing Attempts - Link-Based Phishing

- **Monitor**

- HTTP 302 chains
- Domain reputation scoring
- TLS cert anomalies
- Traffic to newly registered domains (<30 days)

- **Network logs**

- Proxy
- Firewall
- DNS
- Secure web gateway



Phishing Attempts - MFA Fatigue / Push Bombing Phishing

- **Monitor**

- Excessive MFA prompts
- Multiple denied pushes followed by success
- Login success after user “fatigue”

- **Identity telemetry**

- Authentication logs
- Conditional access signals
- Risk-based sign-ins



Malware Infection

- Malware detection is not one-use case. It is a chain of behaviors across the kill-chain:
 - Initial access
 - Execution
 - Persistence
 - Privilege escalation
 - Lateral movement
 - Command & Control (C2)
 - Data exfiltration / Impact



Malware Infection

1. Telemetry

- EDR process creation logs
- Sysmon (Event ID 1, 3, 7, 11, 13)
- AV detections
- Network IDS/NDR
- File integrity monitoring

3. Indicators

- Encoded PowerShell
- Scheduled task creation
- C2 IP/domain communication

2. Detection

- Untrusted binary execution from temp paths
- Registry persistence:
 - HKCU\Software\Microsoft\Windows\CurrentVersion\Run
- Beaconing traffic patterns
- Suspicious DLL loads

Monitoring Malware Infection – Trojan

- **Endpoint**

- New service creation
- Suspicious scheduled tasks
- Autorun registry keys
- Process injection

- **Network**

- Outbound beaconing (periodic connections)
- Traffic to rare IPs/domains
- DNS tunnelling

- **Identity**

- Use of cached credentials
- Unexpected remote logins

- **Logs required**

- Windows Event ID 4688 (process)
- 7045 (service creation)
- Sysmon Event ID 1, 3, 7, 11
- EDR telemetry

Monitoring Malware Infection – Ransomware

- **Endpoint**

- Mass file renaming
- High I/O write activity
- VSS shadow copy deletion
- Execution from temp folders

- **Commands to detect**

- vssadmin delete shadows
- wbadmin delete catalog
- bcdedit recoveryenabled no

- **Network**

- SMB lateral movement
- Domain-wide scanning

- **Identity**

- Admin credential misuse
- PsExec / WMI remote execution

Monitoring Malware Infection – Worm

- **Monitor**

- Unusual port scanning behaviour
- SMB/RPC brute attempts
- Sudden endpoint-to-endpoint traffic

- **Network telemetry**

- East-west firewall logs
- Netflow anomalies

Monitoring Malware Infection – Spyware / Keylogger

- **Monitor**

- Browser injection
- Hooking keyboard APIs
- Credential dump attempts

- **Commands**

- Mimikatz (Open-Source tool that extracts plain passwords, hashes, PIN from memory)
- lsass.exe memory access

- **Logs**

- Sysmon Event ID 10 (process access)
- Defender alerts

Monitoring Malware Infection – Botnet Infection

- **Monitor**

- Beaconsing every X minutes
- Encrypted outbound unknown traffic
- Domain Generation Algorithm (DGA)

- **Network**

- DNS NXDOMAIN spikes
- TLS fingerprint anomalies

Detect suspicious PowerShell

- `Get-WinEvent -LogName "Microsoft-Windows-PowerShell/Operational" | Where-Object {$_.Id -eq 4104}`

```
PS C:\Users\Jeetu> Get-WinEvent -LogName "Microsoft-Windows-PowerShell/Operational" | Where-Object {$_.Id -eq 4104}
```

ProviderName: Microsoft-Windows-PowerShell

TimeCreated	Id	Level	DisplayName	Message
09-Feb-26 7:32:22 PM	4104	Warning		Creating Scriptblock text (4 of 4):...
09-Feb-26 7:32:22 PM	4104	Warning		Creating Scriptblock text (3 of 4):...
09-Feb-26 7:32:22 PM	4104	Warning		Creating Scriptblock text (2 of 4):...
09-Feb-26 7:32:22 PM	4104	Warning		Creating Scriptblock text (1 of 4):...
09-Feb-26 7:32:22 PM	4104	Warning		Creating Scriptblock text (17 of 17):...
09-Feb-26 7:32:22 PM	4104	Warning		Creating Scriptblock text (16 of 17):...
09-Feb-26 7:32:22 PM	4104	Warning		Creating Scriptblock text (15 of 17):...
09-Feb-26 7:32:22 PM	4104	Warning		Creating Scriptblock text (14 of 17):...
09-Feb-26 7:32:22 PM	4104	Warning		Creating Scriptblock text (13 of 17):...
09-Feb-26 7:32:22 PM	4104	Warning		Creating Scriptblock text (12 of 17):...
09-Feb-26 7:32:22 PM	4104	Warning		Creating Scriptblock text (11 of 17):...
09-Feb-26 7:32:22 PM	4104	Warning		Creating Scriptblock text (10 of 17):...
09-Feb-26 7:32:22 PM	4104	Warning		Creating Scriptblock text (9 of 17):...



Detect credential dumping

```
Get-WinEvent -FilterHashtable @{
```

```
LogName='Security'
```

```
ID=4688
```

```
} | Where-Object {$_.Message -match "lsass.exe"}
```

It's good to see ERROR, meaning credentials aren't compromised yet.

```
PS C:\Users\Jeetu> Get-WinEvent -FilterHashtable @{
LogName='Security'
ID=4688
} | where-object {$_.Message -match "lsass.exe"}
Get-WinEvent : No events were found that match the specified selection criteria.
At line:1 char:1
+ Get-WinEvent -FilterHashtable @{
+ ~~~~~~
+ CategoryInfo          : ObjectNotFound: (:) [Get-WinEvent], Exception
+ FullyQualifiedErrorId : NoMatchingEventsFound,Microsoft.PowerShell.Commands.GetWinEventCommand
```



Lateral Movement

- **Telemetry**

- Windows Security logs
- Netflow/NDR
- EDR
- SMB, RDP logs

- **Key Event IDs**

- 4624 (Logon type 3/10)
- 4672 (Special privileges assigned)
- 5140 (SMB share access)
- 4769 (Kerberos service ticket)

- **Detection**

- Admin login across multiple hosts rapidly
- Pass-the-hash/ticket indicators
- Remote service creation
- PsExec/WMI execution

Lateral Movement - Credential Reuse Movement

- Attacker extracts credentials from one machine and authenticates to others.
- **Identity**
 - NTLM authentication spikes
 - Kerberos ticket anomalies
 - Multiple host logins with same account in short time
- **Key logs**
 - Event ID 4624 (logon success)
 - 4625 (failures)
 - 4768, 4769 (Kerberos)
 - 4776 (NTLM auth)
- **Behaviour indicators**
 - Logon Type 3 from workstation → servers
 - Service accounts logging into endpoints



Lateral Movement - Remote Service Execution

- Attacker uses PsExec, SMB, WMI, WinRM, RDP.
- **Endpoint**
 - Service creation remotely
 - Process execution via admin shares
 - WMI process launch
- **Key events**
 - 7045 (service created)
 - 4688 (process execution)
 - 5140 (SMB share access)
 - 4672 (admin privileges assigned)
- **Tools to detect**
 - PsExec
 - WMIC
 - WinRM
 - sc.exe
 - smbexec



Lateral Movement - RDP Lateral Movement

- Remote Desktop used to pivot.
- **Monitor**
 - Multiple RDP logins from non-admin workstation
 - After-hours admin RDP
 - RDP success after multiple failures
- **Logs**
 - Event ID 4624 Logon Type 10
 - 4778 session reconnect
 - 4779 disconnect



Lateral Movement - SMB / Admin Share Abuse

- Access to C\$, ADMIN\$, IPC\$ shares.
- **Network**
 - Lateral SMB traffic
 - Access from non-admin machines
- **Events**
 - 5140 share accessed
 - 5145 detailed share access
 - Sysmon Event ID 3 network connection



Lateral Movement - PowerShell Remoting Movement

- Lateral movement using PowerShell sessions.
- **Monitor**
 - Encoded commands
 - New PS sessions across hosts
 - Remote script execution
- **Logs**
 - Event ID 4103, 4104
 - WinRM logs
 - Sysmon Event ID 1



Lateral Movement - Domain Controller Targeting

- Movement toward DC to escalate privileges.
- **Monitor**
 - Workstation authenticating to **DC via SMB/RDP**
 - Kerberos service ticket anomalies
 - DCSync attempts
- **Critical events**
 - 4662 directory replication
 - 4769 ticket requests
 - 4672 privileged logons



Lateral Movement - Scheduled Tasks

- **Monitor**

- Tasks created remotely
- Tasks executed under SYSTEM

- **Events**

- 4698 task created
- 4702 task modified

Living-off-the-Land Movement

- Use of built-in tools.
- **Monitor execution of:**
 - net use
 - wmic
 - powershell
 - cmd
 - schtasks
 - rundll32
- **Logs**
 - Event ID 4688
 - Sysmon process tracking



Detect remote service creation

Untitled1.ps1* X

```
1 cls
2 Get-WinEvent -FilterHashtable @{
3   LogName='System'
4   ID=7045
5 } -MaxEvents 1 | ft -AutoSize -wrap
6
```

ProviderName: Service Control Manager

TimeCreated	Id	Level	DisplayName	Message
-----	--	-----	-----	-----
03-Feb-26 10:10:42 PM	7045	Information		A service was installed in the system.
				Service Name: HP One Agent Service
				Service File Name: "C:\Program Files\HP\HP One Agent\hp-one-agent-service.exe"
				Service Type: user mode service
				Service Start Type: auto start
				Service Account: LocalSystem



Brute Force Attempts

- **Telemetry**

- AD/Entra sign-in logs
- VPN logs
- SSH logs
- Firewall authentication logs

- **Key events**

- 4625 failed logons
- Azure risky sign-in alerts

- **Detection**

- Multiple failures followed by success
- Same account from multiple IPs
- Password spray pattern

Brute Force Detection – Attack Patterns

- Password spraying (low-and-slow across many users)
- Credential stuffing (known username/password combos)
- Traditional brute force (many attempts → one account)
- MFA brute force / OTP guessing
- Service account targeting
- VPN / RDP brute forcing



Brute Force - Active Directory Password

- Attacker repeatedly attempts logins against domain accounts.

- **Identity telemetry**

- Event ID 4625 (failed logon)
- Event ID 4771 (Kerberos pre-auth failure)
- Event ID 4776 (NTLM failure)

- **Behaviour patterns**

- Same IP targeting multiple users
- Single user receiving repeated failures
- Failures followed by success

- **Indicators**

- Logon Type 3 (network)
- Logon Type 10 (RDP)
- Workstation name anomalies



Brute Force - Password Spraying

- One password tried across many accounts to avoid logout.
- **Monitor**
 - Same IP → many users → single attempt each
 - Spike in Kerberos failures
 - No account lockouts but high failure count

Password Spraying Script

```
$TimeWindowMinutes = 60
$UserThreshold = 10 # number of unique users targeted
$FailThreshold = 20 # total failures from source IP
$StartTime = (Get-Date).AddMinutes(-$TimeWindowMinutes)
Write-Host "Analyzing failed logons since $StartTime ..." -ForegroundColor Cyan

# Pull failed login events
$Events = Get-WinEvent -FilterHashtable @{
    LogName='Security'
    ID=4625
    StartTime=$StartTime
}

$parsed = foreach ($event in $Events) {
    $xml = [xml]$event.ToXml()
    $ip = ($xml.Event.EventData.Data | Where-Object {$_.Name -eq "IpAddress"}).'#text'
    $user = ($xml.Event.EventData.Data | Where-Object {$_.Name -eq "TargetUserName"}).'#text'
    $logonType = ($xml.Event.EventData.Data | Where-Object {$_.Name -eq "LogonType"}).'#text'

    # Focus on network & RDP attempts
    if ($logonType -in ("3","10") -and $ip -ne "-" -and $user -ne "SYSTEM") {
        [PSCustomObject]@{
            TimeCreated = $event.TimeCreated
            SourceIP = $ip
            Username = $user
            LogonType = $logonType
        }
    }
}

# Group by Source IP
$analysis = $parsed | Group-Object SourceIP | ForEach-Object {
    $uniqueUsers = ($_.Group.Username | Select-Object -Unique).Count
    $totalFails = $_.Count
    if ($uniqueUsers -ge $UserThreshold -and $totalFails -ge $FailThreshold) {
        [PSCustomObject]@{
            SourceIP = $_.Name
            UniqueUsers = $uniqueUsers
            TotalFailures = $totalFails
            FirstSeen = ($_.Group.TimeCreated | Sort-Object)[0]
            LastSeen = ($_.Group.TimeCreated | Sort-Object)[-1]
        }
    }
}

if ($analysis) {
    Write-Host "`nPotential Password Spraying Detected:" -ForegroundColor Red
    $analysis | Sort-Object TotalFailures -Descending | Format-Table -AutoSize
} else {
    Write-Host "`nNo password spraying patterns detected." -ForegroundColor Green
}
```

Analyzing failed logons since 02/09/2026 19:19:21 ...

No password spraying patterns detected.

PS C:\Users\Jeetu>



Brute Force - VPN

- Attacker attempts remote access via VPN gateway.

- **Network**

- VPN authentication failures
- Geo anomalies
- Repeated connection attempts

- **Indicators**

- Login attempts from TOR/cloud hosts
- Failures across many usernames

- **Logs**

- VPN concentrator logs
- Firewall auth logs



Brute Force - RDP

- Internet-facing RDP targeted.
- **Monitor**
 - Event ID 4625 Logon Type 10
 - Multiple failures from same IP
 - Success after failures
- **Network**
 - Port 3389 scanning
 - External source login attempts

Brute Force - Cloud Identity

- Attacker attempts cloud login.
- **Identity**
 - Sign-in failure reasons
 - Risky login flags
 - Impossible travel
- **Indicators**
 - Legacy authentication usage
 - MFA prompts triggered repeatedly



Brute Force - MFA

- Repeated OTP attempts.
- **Monitor**
 - Multiple OTP failures
 - Push denial patterns
 - MFA reset attempts

Privilege Escalation – Local Admin

- Malware or attacker elevates privileges on a compromised machine.
- **Endpoint**
 - UAC bypass attempts
 - New admin group membership
 - Token elevation events
 - Process launched as Administrator unexpectedly
- **Key logs**
 - Event ID 4672 (special privileges assigned)
 - 4688 (process execution elevated)
 - 4732 (user added to local admin group)
 - Sysmon process injection indicators



Privilege Escalation – Domain Admin

- Standard domain user escalates to domain admin.

- **Identity**

- Changes in privileged AD groups:
 - Domain Admins
 - Enterprise Admins
 - Administrators
 - SIDHistory manipulation
 - AdminCount attribute changes

- **Key logs**

- 4728 / 4729 → Added/removed from privileged groups
 - 4735 → Group modified
 - 4662 → Directory object access



Privilege Escalation – Service Account Abuse

- Weak service account credentials leveraged.
- **Monitor**
 - Service accounts logging interactively
 - Service account authentication from workstation
 - Password change anomalies
- **Indicators**
 - Kerberos ticket requests unusual for service accounts
 - NTLM fallback



Privilege Escalation – GPO Manipulation

- Attacker modifies GPO to grant admin rights.
- **Monitor**
 - GPO edits
 - Script deployment via GPO
 - Privilege assignment changes
- **Logs**
 - 5136 directory changes
 - Group Policy operational logs



Detect admin group membership changes

Untitled1.ps1* X

```
1 cls
2 Get-WinEvent -FilterHashtable @{
3   LogName='Security'
4   ID=4728,4732
5 } -MaxEvents 1 | ft -AutoSize -wrap
6
```

2/7/2026 9:37:48 AM 4728 Information

A member was added to a security-enabled global group.

Subject:

Security ID: S-1-5-21-3094736813-2960354787-1534546388-500
Account Name: Administrator
Account Domain: TRAINING
Logon ID: 0x4F7F9

Member:

Security ID: S-1-5-21-3094736813-2960354787-1534546388-1104
Account Name: cn=jeetu,CN=Users,DC=training,DC=com

Group:

Security ID: S-1-5-21-3094736813-2960354787-1534546388-520
Group Name: Group Policy Creator Owners
Group Domain: TRAINING

Additional Information:

Privileges: -

Indicators of Communication Trackers (C2)

- **Telemetry**

- DNS logs
- Proxy logs
- Netflow
- TLS inspection logs

- **Detection**

- Beaconsing periodic traffic
- Domain generation algorithms (DGA)
- Suspicious JA3 TLS fingerprints
- DNS tunnelling patterns

- **Indicators**

- High entropy domains
- Repeated TXT record queries
- Known C2 IP feeds

Web Attacks

- **Telemetry**

- WAF logs
- IIS/Apache/Nginx access logs
- Application logs
- Reverse proxy logs

- **Detection**

- SQL injection patterns:
 - ' OR 1=1
- XSS attempts:
 - <script>
- Path traversal:
 - ../../etc/passwd
- File upload abuse



Web Attacks – SQL Injection

- Attacker injects malicious SQL via form fields/URL parameters.
- **Application**
 - SQL errors in responses
 - Unusual query patterns
 - Boolean/time-based queries
- **Web logs**
 - ' OR 1=1
 - UNION SELECT
 - SLEEP(), WAITFOR DELA



Web Attacks – Cross-Site Scripting (XSS)

- Script injected into pages to steal sessions.
- **Monitor**
 - `<script>` tags in input
 - Encoded payloads
 - DOM-based injection attempts
- **Indicators**
 - Session hijacking behavior
 - Unexpected redirects

Web Attacks – Cross-Site Request Forgery (CSRF)

- Victim browser performs unauthorized actions.
- **Monitor**
 - Requests without CSRF tokens
 - Actions from unusual referrer domains
 - Account changes without login

Web Attacks – DDoS / Layer 7 Flood

- Application resource exhaustion.
- **Monitor**
 - Request spikes
 - Same user agent patterns
 - Geo anomalies

DNS Attacks

- **Telemetry**

- DNS resolver logs
- Passive DNS
- Endpoint DNS client logs

- **Detection**

- DNS tunnelling
- NXDOMAIN bursts
- Requests to malicious domains
- DNS exfiltration (large TXT records)



DNS Attacks - DNS Tunnelling

- Malware uses DNS queries to communicate with C2.

- **Network**

- High volume DNS queries
- Long subdomain strings
- Encoded base64-like patterns
- High entropy domains

- **Indicators**

- TXT record queries
- Repeated requests to single domain
- Unusual query intervals

- **Telemetry**

- DNS resolver logs
- Passive DNS
- NetFlow

DNS Attacks - DNS Data Exfiltration

- Data embedded in DNS requests.
- **Monitor**
 - Outbound queries with encoded payload
 - Unusually long FQDN queries
 - Large DNS request size
- **Behaviour**
 - Periodic small DNS queries
 - No corresponding HTTP traffic



DNS Attacks – DNS spoofing / Cache poisoning

- Malicious redirection.
- **Monitor**
 - DNS response mismatches
 - IP change for known domains
 - Unexpected internal resolution
- **Indicators**
 - Users redirected to phishing infra
 - TLS certificate mismatch after DNS resolution



DNS Attacks – Fast-Flux DNS

- Rapid IP rotation to evade blocking.
- **Monitor**
 - Frequent IP changes
 - Short TTL values
 - Multiple A records

Get-WinEvent -LogName "DNS Server"

TimeCreated	Id	Level	DisplayName	Message
9/9/2026 6:55:58 AM	4	Information		The DNS server has finished the background loading and signing of zones. All zones are now available
9/9/2026 6:55:57 AM	2	Information		The DNS server has started.
9/9/2026 6:55:57 AM	7648	Information		The DNS server has detected that it is no longer the Key Master for zone training.com. The Key Master
9/9/2026 6:55:57 AM	769	Information		The DNS server has loaded the zone training.com from file NULL on server dc.training.com. [virtualiza
9/9/2026 6:55:57 AM	769	Information		The DNS server has loaded the zone _msdcs.training.com from file NULL on server dc.training.com. [vir
9/9/2026 6:55:39 AM	4013	Warning		The DNS server is waiting for Active Directory Domain Services (AD DS) to signal that the initial syn
9/9/2026 6:55:34 AM	7693	Information		The XfrScopeOptionValue has been set to 65433. This option ID will be used to communicate the scope
7/7/2026 9:09:02 AM	4500	Information		The DNS Application Directory Partition ForestDnsZones.training.com was created. The distinguished na
7/7/2026 9:09:02 AM	4500	Information		The DNS Application Directory Partition DomainDnsZones.training.com was created. The distinguished na
7/7/2026 9:01:54 AM	7648	Information		The DNS server has detected that it is no longer the Key Master for zone training.com. The Key Master
7/7/2026 9:00:53 AM	2	Information		The DNS server has started.
7/7/2026 9:00:52 AM	4	Information		The DNS server has finished the background loading and signing of zones. All zones are now available
7/7/2026 9:00:52 AM	769	Information		The DNS server has loaded the zone training.com from file training.com.dns on server dc.training.com.
7/7/2026 9:00:52 AM	769	Information		The DNS server has loaded the zone _msdcs.training.com from file _msdcs.training.com.dns on server dc
7/7/2026 9:00:34 AM	4013	Warning		The DNS server is waiting for Active Directory Domain Services (AD DS) to signal that the initial syn
7/7/2026 9:00:29 AM	7693	Information		The XfrScopeOptionValue has been set to 65433. This option ID will be used to communicate the scope
7/7/2026 8:59:58 AM	3	Information		The DNS server has shut down.
7/7/2026 8:59:58 AM	3150	Information		The DNS server wrote version 2 of zone training.com to file training.com.dns.
7/7/2026 8:59:39 AM	3150	Information		The DNS server wrote version 1 of zone _msdcs.training.com to file _msdcs.training.com.dns.
7/7/2026 8:59:39 AM	3150	Information		The DNS server wrote version 1 of zone training.com to file training.com.dns.
7/7/2026 8:59:39 AM	2631	Information		The DNS server successfully autoconfigured: ...
7/7/2026 8:59:08 AM	2	Information		The DNS server has started.
7/7/2026 8:59:08 AM	708	Information		The DNS server did not detect any zones of either primary or secondary type during initialization. It
7/7/2026 8:59:08 AM	414	Warning		The DNS server computer currently does not have a DNS domain name. Its DNS name is a single-label ho
7/7/2026 8:59:08 AM	7693	Information		The XfrScopeOptionValue has been set to 65433. This option ID will be used to communicate the scope



Network scanning /Port scanning / Vulnerability Scanning

Attack objectives

- Discover live hosts
- Identify open ports/services
- Fingerprint OS & applications
- Find exploitable vulnerabilities
- Map internal network for lateral movement

Network scanning – Host Discovery

- Attacker enumerates live systems.

- **Network**

- ICMP echo spikes
- Sequential subnet probing
- Broadcast discovery attempts

- **Indicators**

- One host → many IPs
- Short time window

- **Telemetry**

- Firewall logs
- IDS alerts
- NetFlow



Network scanning – Port Scanning

- Attacker probes ports for services.
- **Types**
 - SYN scan
 - TCP connect scan
 - UDP scan
 - Stealth scan (FIN/XMAS/NULL)
- **Network**
 - Multiple port attempts from single source
 - Horizontal scanning (same port → many hosts)
 - Vertical scanning (many ports → single host)
- **Indicators**
 - Failed connection bursts
 - SYN packets without handshake completion



Network scanning – Vulnerability Scanning

- Nessus, OpenVAS, Qualys OR attacker tools.

- **Network**

- Sequential scanning patterns
- HTTP enumeration
- SMB enumeration
- SNMP queries

- **Endpoint**

- Authentication failures
- Service probe attempts

- **Indicators**

- Known scanner signatures
- Scripted user agents



Network scanning – Active Directory Enumeration

- Attacker mapping AD structure.
- **Monitor**
 - LDAP queries spike
 - Kerberos service enumeration
 - Domain user listing
- **Logs**
 - 4662 directory access
 - 4769 ticket requests



LDAP enumeration

```
cls
Get-WinEvent -FilterHashtable @{
  LogName='Security'
  ID=4662
} -MaxEvents 1 | ft -AutoSize -Wrap
```

```
Created          Id LevelDisplayName Message
-----
10/26 8:10:27 AM 4662 Information An operation was performed on an object.

Subject :
  Security ID:      S-1-5-18
  Account Name:     DC$
  Account Domain:   TRAINING
  Logon ID:         0x13851C

Object:
  Object Server:    DS
  Object Type:      %{19195a5b-6da0-11d0-afd3-00c04fd930c9}
  Object Name:      %{d4fb8154-b374-4949-a349-b8c2a1a70259}
  Handle ID:       0x0

Operation:
  Operation Type:   Object Access
  Accesses:        Control Access

Access Mask:       0x100
Properties:        Control Access
                  {1131f6aa-9c07-11d1-f79f-00c04fc2dcd2}
                  {19195a5b-6da0-11d0-afd3-00c04fd930c9}
```

Attempted Exploits

- **Telemetry**

- IDS/IPS alerts
- EDR exploit detection
- Patch management logs
- WAF

- **Detection**

- Known CVE exploit signatures
- Buffer overflow attempts
- Shellcode patterns
- Exploit kit traffic

- **Correlation**

- Exploit attempt → process crash → privilege escalation attempt

Attempted Exploits - Endpoint Exploit Attempts

Detect attempts to exploit OS, applications, or memory vulnerabilities on endpoints.

- **Client-Server Communication**

- Unusual inbound connections to workstations
- SMB/RPC exploitation attempts

- **Data Flow**

- Malformed packets
- Suspicious payload delivery

- **Ports / Protocols / Services**

- RDP (3389)
- SMB (445)
- RPC (135)
- WinRM (5985/5986)
- DEP/ASLR violations

- **Systems & Applications**

- Windows OS
- Office apps
- Browsers
- Java runtime

- **Security Controls**

- EDR exploit protection
- ASR rules

Attempted Exploits - Web Application Exploit Attempts

- Identify attackers exploiting web vulnerabilities (SQLi, RCE, LFI, deserialization, etc.).

- **Client-Server**

- HTTP requests with abnormal parameters

- **Data Flow**

- Encoded payloads
 - Long URIs
 - Command injection strings

- **Ports / Protocols**

- HTTP/HTTPS (80/443)
 - API endpoints

Applications

IIS / Apache / Nginx

Web APIs

Application servers

Controls

WAF

Reverse proxy

API gateways

Attempted Exploits - Network-Based Exploit Attempts

- Detect vulnerability exploitation at network layer.
- **Client-Server**
 - Exploit traffic from external IPs
- **Data Flow**
 - Known exploit signatures
- **Ports**
 - 21, 22, 80, 443, 445, 1433, 3389
- **Systems**
 - Public servers
 - Firewalls
 - Load balancers
- **Controls**
 - IDS/IPS
 - Network TAP
 - Firewall



Attempted Exploits - Privilege Escalation Exploit Attempts

- Detect attempts to exploit local vulnerabilities to gain admin/system.
- **Endpoint Behaviour**
 - Token manipulation
 - Service abuse
- **Data Flow**
 - Kernel exploit payloads
- **Systems**
 - Domain Controllers
 - Critical servers
- **Controls**
 - EDR
 - Windows audit policy







That's all Folks!